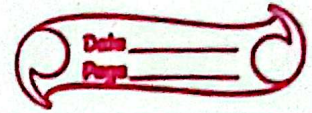


Chapter - 3



Q1) What are common legal and moral issues faced by a software engineers? Explain with examples.

Ans: ① Privacy Violation Problem:

Legal & Moral Legal Looking at or sharing user's private information without their permission.

Example:

A software engineer reads people's private chats from a social media app without their knowledge.
→ This is illegal & wrong.

② Copying code without permission:

Legal Issue - Using someone else's code or software without asking or paying.

Example:

A developer copies a paid software and use it in their own project.

→ This is called software piracy and is illegal.

③ Hacking:

Legal Issue → Breaking into other people's systems or data.

Example:

An engineer hacks into a competitor company to steal their files. This is a cybercrime.

④ Making viruses or Ransomware:

Moral & Legal Creating harmful programs that damage or lock user's files.

Example:

A person creates ransomware that asks for money to unlock a user's computer. This is very dangerous & illegal.

(5) Unfair or Biased Software:

Moral Issue!

Software that treats people unfairly, like preferring men over women.

Example:

An AI program selects only male candidates for jobs because of biased data.

→ This is unethical and unfair.

(6) Plagiarism:

Moral Issue

Taking someone's code or work and saying it's yours.

Example: Student submit a project using online code but says they wrote it. This is dishonest.

(7) Unsafe Software:

Moral & legal issues

Making software that has bugs and can cause serious problems.

Example: A bug in hospital software gives patients the wrong medicine dose.

→ This can harm people & lead to legal issues.

Q.19
Q.20

Define philosophical and professional ethics. Explain the descriptive & normative claims.

Ans - Philosophical ethics: It is the study of what is right and wrong, and why something is considered good or bad.

Example: Should a programmer build a robot that replaces human workers? philosophical ethics helps think about whether it's morally right or wrong.

Professional ethics:

They are the rules and values that guide how people should behave at work - especially in fields like software engineering, medicine, teaching, etc.

Example:

A software engineer must not steal code, must protect user data, and be honest with clients - these are part of professional ethics.

Descriptive Claim:

- Descriptive claim tells what is really happening.
- They show facts.
- They talk about what people do, believe or say in real life.
- They do not judge if something is good or bad.
- Just like telling a story or reporting what's happening.

Example 1:

"Many people lie on their resumes."

→ This is a fact. It's not saying lying is right or wrong.

Example 2:

"In some countries, people eat dinner very late."

→ It simply describes what people do. No opinion.

→ Descriptive: Describes only, no judging.

② Normative Claims:

→ "What should be?"

→ Normative claims tell us what is right or wrong.

→ They talk about what people should or should not do.

→ It is about morals, rules and opinions.

Example 1:

"People should always tell the truth."

→ This says what is the right thing to do.

Example 2:

"Children should respect their elders."

→ This is moral opinion about what is good behavior.

→ Normative: Talks about rules, values or what is right/wrong.

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③ What is Ethical Relativism? Compare and contrast "Utilitarianism and Deontological" theory in the light of ethical relativism. Give examples to support your answer.

Ans Ethical Relativism is the theory that holds that morality is relative to the norms of one's culture. i.e. whether an action is right or wrong depends on the moral norms of the society in which it is practiced.

Utilitarianism: is an ethical theory that says an action is right if it brings the most happiness to the most no. of people.

Deontological: says an action is right or wrong based on moral rules or duties, not on the result.

Utilitarianism	Deontological
① Focuses on what action gives the most good / happiness to most people.	① Focuses on whether the action itself is right or wrong, based on rules.
② It is consequence-based (result matters most).	② It is Rule-based (intention & duty matter most).
③ Example: If lying saves lives, it is acceptable.	③ Example: Lying is always wrong, even if it saves lives.

(~~write~~)

④ Depends on the result.
(right or wrong) - if result
is good, action is good.

④ Fixed - some actions are
always right or always
wrong.

⑤ Matches relativism -
moral actions can change
based on culture and
situation.

⑤ Conflict with relativism -
believes in universal moral
rules.

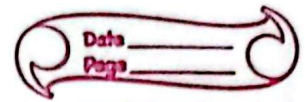
⑥ Flexible - respects cultural
values and norms.
example in culture:-

⑥ Not flexible - some moral
rules for all societies.

⑦ Some culture accept
sacrifice for the greater
good / its utilitarian
thinking.

⑦ Example in culture:-
Some cultures may allow
things that break universal
rules (against deontology).

chapter-4



2b) What do you mean by a professional code of ethics? explain the code of the Nepal Engineering Council.

Ans: Professional code of ethics is a set of guiding principles that outlines the ethical standards and expectations for behavior within a profession or organization.

The code of ^{ethics of} the Nepal Engineering Council are:

① Discipline and Honesty:

→ Engineers should work in a disciplined and honest way. Their work should respect professional values and dignity.

② Politeness and Confidentially:

→ Engineers should be polite with customers.

→ Keep client information secret unless they allow sharing.

→ But if the law demands, engineers must share information with authorities.

③ Non-discrimination:

- No discrimination should be made against customer based on religion, caste, gender, etc while applying professional knowledge and skills.

④ Professional work:

- Engineers should only work within their own subject or expertise.
- If something is outside their field, they should refer it to a qualified expert.

⑤ No Harmful Acts for the profession:

- Do not take any illegal or unfair financial benefits.
- Only accept salary, allowance or legal benefits for your service.

⑥ Personal Responsibility:

- All the individuals will be personally responsible for all works performed in connection with his/her engineering profession.

⑦ State name, designation and registration no:

- When signing any engineering work (like design or map), mention your:
(name, designation & ^{NEC} registration number)
should be stated in a clear.

⑧ No Unnecessary Publicity or Ads:

- No publicity or advertisement shall be made so as to cause unnecessary effect upon the customers.

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a) What are Professional Rights and Responsibilities?
How can you avoid the whistle blowing in the organization?

Ans) Professional rights are the rights that allow a person to do their job properly, fairly, and with respect.

Professional responsibilities are the duties and tasks a person must do in their job, honestly and with care.

We can avoid the whistle blowing in the organization by:-

① Strong corporate ethical culture!

Ans → There should include a clear commitment to ethical behavior, starting at the higher levels of management, and mandatory ethics training for all employees.
→ All managers must set the tone for the ethical behavior of their employee.

② Clear lines of communication!

Ans → There should be clear lines of communication with a corporation.
→ This openness gives an employee who feels that there is something that must be fixed a clear path to air his concerns.

③ Meaningful access to high-level managers:

- Third, ~~there should~~ all employees must have meaningful access to high-level managers in order to bring their concerns forward.
- This access must come with a guarantee that there will be no retaliation.
- Rather, employees willing to come forward should be rewarded for their commitment to fostering the ethical behavior of the company.

④ Admit mistakes:

- Finally, there should be willingness on the part of management to admit mistakes, publicly if necessary.
- This attitude will set the stage for ethical behavior by all employees.

⑤ Is whistleblowing a professional act? What are the conditions required to blow a whistle?

Ans: Yes, whistleblowing ~~is~~ is a professional act, but only if done with right intention and under the right conditions. It is considered professional when an employee reports serious

wrong doing to protect the public, the company or the environment - not for personal revenge or gain.

The conditions required to blow a whistle:-

① Need:

- ~~There must~~ means there must be serious harm or risk
- Only blow the whistle if something truly wrong or harmful is happening. Don't report every small issue unless it adds up to a big problem.

② Proximity:

- It means you must know the issue clearly.
- You must have first-hand knowledge (not rumour)
- You should understand the issue well and be in the right position to report it.

③ Capability:

- It means you must be able to stop the harm.
- You should have a real chance to solve the problem or reach someone who can. Don't risk your job if you can't make a difference.

④ Last Resort:

- It means all other options must be tried.
- Blow the whistle only if other ways (like reporting to a manager) have failed.
- It should be the last option, not the first.

- Note:

- whistleblowing is not easy because
- people may see it as disloyalty.
- It can create problems in the workplace.
- That's why it should be used carefully and ethically.

Q) Define the term conflict of interest and whistleblowing? Explain 8 values in design.

Ans: Conflict of Interest (COI) is a situation in which a person or organization is involved in multiple interests, financial interest, ~~or~~ or otherwise, one of which could possibly corrupt the motivation of the individual or organization.

Whistleblowing:

It is the process of exposing any kind of information or activity that is deemed illegal, dishonest, or not correct within an organization that is either private or public.

8 values in design are:

Types of computer crime:

→ Computer crime is an act performed by a knowledge computer user, sometimes referred as a hacker that illegally brocoses or steals a company's or individual's private information.

① Types:-

① Cybercrime:

- It is any illegal activity that involves a computer ~~not~~ and a network.
- It includes crimes like hacking, stealing data and online scams.

Example! A hacker steals someone's bank password and takes their money.

② Software piracy:

- Copying software without permission is called software piracy.
- It's illegal to share licensed software with others.

Example!- You buy a game & send a copy to your friend _____ for free.

③ Computer Fraud:

- Using the Internet or computers to cheat people, companies, or governments out of money or services.

→ Methods include phishing, social engineering and virus attacks. Eg: A fake email asks you to pay a bill that doesn't exist.

③ Digital Forgery:

→ Changing or faking digital documents or images to deceive others.

Example: Editing photos or fake digital signatures

④ Phishing:

→ Sending false emails or links to trick people into giving away personal information like passwords or bank details.

Example: An email says "Your bank account

⑤ Unauthorized access: Ps locked" and asks for your

→ Access PIN.

⑥ Unauthorized access:

→ Accessing someone's system, account, or data without permission.

→ It is also called hacking if done with bad Internet.

Example: Guessing someone's facebook password & logging in.

⑥ Hacking:

→ Breaking into computers or networks to steal data, test security, or cause harm.

— white hat hackers do it ethically; black hats do it

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and Identity theft.

Example! Sending repeated mean messages to someone on Instagram.

(11) Spams!

→ Sending large no. of unwanted emails or messages usually for ads or scam.

Example! Your inbox is full of ads for fake medicine.

(12) Malicious programs (malware)

→ Software made to harm computer.

It includes:- viruses, worms, Trojans, spyware / Adware and Root kits.

Example! A game download secretly installs a program that steals your data.

(13) Cyber Terrorism!

→ Using computers to cause harm, fear, or damage for political reasons.

→ Targets include banks, power plants, etc.

Example! Hacking a power station to cause a blackout.

illegally.

Example! A hacker enters a school system and changes student grades.

⑦ Cracking:

→ Changing software to remove protections like license checks or ads.

→ Cracks are used to use software without buying it.

Example! Using a keygen to activate a paid software for free.

⑧ Denial of Service (DOS):

→ Attacking a system to make it unavailable.

→ Done by flooding it with too many requests.

example! Flooding a shopping website with fake traffic so real users can't open it.

⑨ Online Pornography:

→ Porn available on the internet in forms like images, videos, webcam or stories.

Example! Watching porn videos on adult websites.

⑩ Online Harassment:

→ Bullying or threatening someone online.

→ Can include hate speech, sexual content, cyberbullying.